



Introduction to Programming 1

(with Python)

Sudharshan Welihinda(weli@iit.ac.lk)
Senior Lecturer, Department of Computing
Informatics Institute of Technology

Check your progress ...

Problem 1: Write an algorithm to read two numbers and find their sum.

Before writing an algorithm for a problem, one should find out what is/are the inputs to the algorithm, the process and what is/are expected outputs after running the algorithm.

Inputs : First number num1, Second number num2.

Process : $\text{Sum} = \text{num } 1 + \text{num}2$

Output: Sum of the two numbers.

Algorithm:

Step1: Start

Step2: Read\input the first num1.

Step3: Read\input the second num2.

Step4: Sum $\text{num}1 + \text{num}2$ // calculation of sum

Step5: Print/Display/Output Sum

Step6: End

Algorithm to add two number

Input → **Algorithm** → **Output**

num1 = 10
num2 = 10

1. Read values num1 and num2.
2. Add num1 and num2 and assign the result to sum.
3. $\text{Sum} = \text{num1} + \text{num2}$
4. Display sum

Sum = 20

Check your progress ...

Problem 2: Write an algorithm convert temperature Fahrenheit to Celsius

Inputs :. Temperature in Fahrenheit

Process : $C = 5/9 * (F - 32)$

Output :. Temperature in Celsius

Algorithm:

Step1: Start

Step 2: Read Temperature in Fahrenheit F

Step 3: $C = 5/9 * (F - 32)$

Step 4: Print Temperature in Celsius: C

Step5: End

Check your progress ...

Problem 3: Write an algorithm to find the greater number between two numbers

Inputs : Two numbers A and B.

Process : If $A > B$ then Highest = A, If $B > A$ then Highest = B

Output : Highest, between two numbers A and B.

Algorithm:

Step1: Start

Step2: Read/input A and B

Step3: If A greater than B then Highest = A

Step4: if B greater than A then Highest = B

Step5: Print Highest

Step6: End

Check your progress ...

Problem 3: Write an algorithm to find the largest number in a given list of 5 positive numbers.

Inputs : List of numbers. This list must contain at least one number.

Process : Start with assuming the max number is equal to 0. For each number x in the list, compare it with max and if x is larger than max then assign x to max.

Output : Largest number in the list.

Algorithm:

- Step1: Start
- Step2: Read/input list of numbers.
- Step3: Set **max** = 0
- Step4: Read a number (x) from the list.
- Step5: Compare x with **max**. If x is larger than **max**, then set **max** = x .
- Step6: Repeat Step 4 through 5, ____ times (OR until end of list.)
- Step7: Output **max**.
- Step8: End.

Check your progress ...

Problem 5: Write an algorithm to find whether a number given by a user is Divisible by 2 or not.

Inputs : Number x

Process : Divide x by 2. If there the remainder is 0 then divisible by 2. Otherwise not divisible by 2.

Output: x is divisible by 2 or not

Algorithm:

- Step 1: Start
- Step 2: Read/input x
- Step 3: Divide x by 2 and get the remainder
- Step 4: if remainder equal zero then x is divisible by 2
- Step 5: if remainder is NOT equal zero then x is not divisible by 2
- Step 5: Stop.

Check your progress ...

Algorithm & Flowchart to find the largest of three numbers

Algorithm

Step-1 Start

Step-2 Read three numbers say num1,num2, num3

Step-3 if num1>num2 then go to step-5

Step-4 IF num2>num3 THEN print num2 is largest ELSE print num3 is largest ENDIF GO TO Step-6

Step-5 IF num1>num3 THEN print num1 is largest ELSE print num3 is largest ENDIF

Step-6 Stop

Check your progress ...

Problem 6: Write an algorithm to find the result of equation: $f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Inputs to the algorithm: Number x

Expected output: Value of $f(x)$, F.

Algorithm:

- Step1: Start
- Step2: Read/input x
- Step3: If X Less than zero then $F = -X$
- Step4: if X greater than or equal zero then $F = X$
- Step5: Print F
- Step6: End

Summary

What have you learned?

- Approaches in problem solving
- Identify Input, Process, Output (IPO) from a problem statement
- Planning the Algorithm